U23AI016
BDA
Assignment 10

Exercise 3: Time-Series & Subplots
Goal: Practice the "Small Multiples" design dimensionality principle.

- Dataset: Generate a synthetic dataset or use seaborn.load_dataset('flights').
- Tasks:

1. Using plt.subplots(2, 1), create two stacked charts.
2. Top Chart: A line graph showing the trend of passengers over time.
3. Bottom Chart: A bar graph showing the monthly average passengers.
4. Requirement: Ensure both charts share the same X-axis (Time) and have clear annotations.

Ans :

Code :

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Load dataset
df = sns.load_dataset('flights')

# Create subplots
fig, axes = plt.subplots(2, 1, figsize=(10, 8))

# Top chart
axes[0].plot(df['year'], df['passengers'])
axes[0].set_title("Passengers Over Time")
axes[0].set_xlabel("Year")
axes[0].set_ylabel("Passengers")

# Bottom chart
monthly_avg = df.groupby('month')['passengers'].mean()

axes[1].bar(monthly_avg.index, monthly_avg.values)
axes[1].set_title("Average Monthly Passengers")
axes[1].set_xlabel("Month")
axes[1].set_ylabel("Passengers")

# Save plot
plt.tight_layout()
plt.savefig("exercise3_subplot.png")
```

print("Plot saved successfully!")

```
GNU nano 7.2                                        exercise3.py *
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
# Load dataset
df = sns.load_dataset('flights')

# Create subplots
fig, axes = plt.subplots(2, 1, figsize=(10, 8))

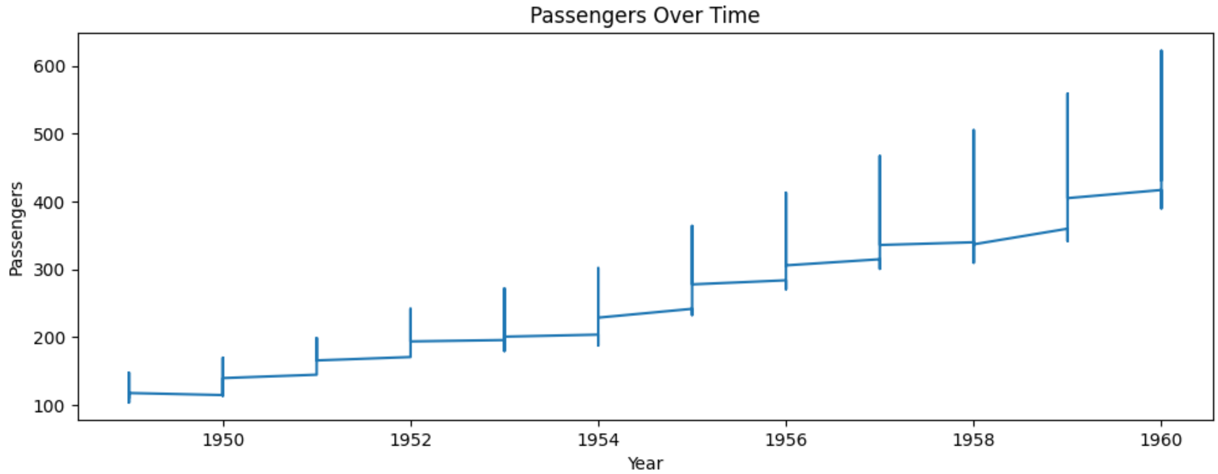
# Top chart
axes[0].plot(df['year'], df['passengers'])
axes[0].set_title("Passengers Over Time")
axes[0].set_xlabel("Year")
axes[0].set_ylabel("Passengers")

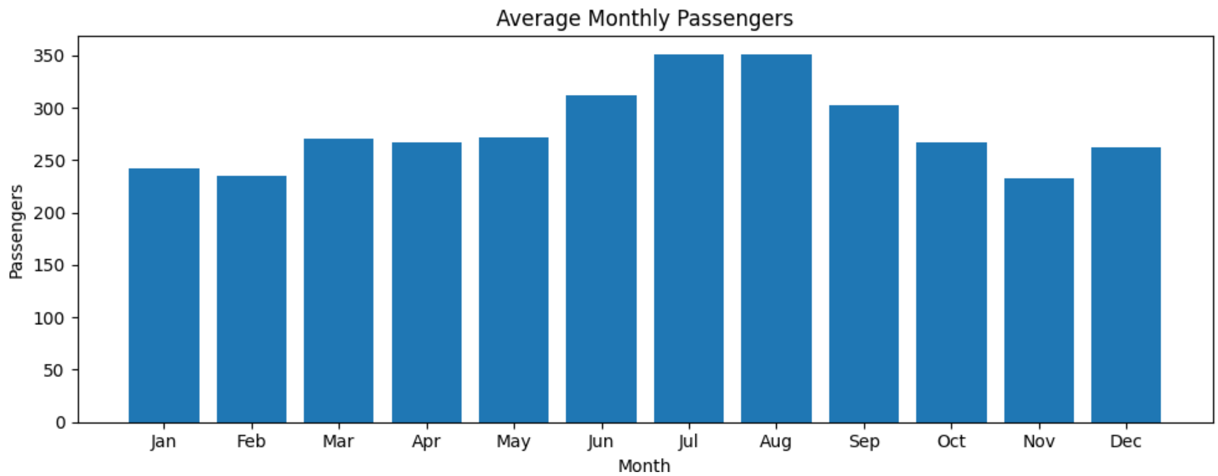
# Bottom chart
monthly_avg = df.groupby('month')['passengers'].mean()

axes[1].bar(monthly_avg.index, monthly_avg.values)
axes[1].set_title("Average Monthly Passengers")
axes[1].set_xlabel("Month")
axes[1].set_ylabel("Passengers")
# Save plot
plt.tight_layout()
plt.savefig("exercise3_subplot.png")
print("Plot saved successfully!")
```

Output :

```
(myenv) user@DESKTOP-0OTJMH0:~/bda_lab/Lab_10$ python3 exercise3.py
Plot saved successfully!
(myenv) user@DESKTOP-0OTJMH0:~/bda_lab/Lab_10$
```





Exercise 4: Multivariate Analysis (Heatmaps)
Goal: Understand correlation matrices, similar to identifying patterns in a Modern Data Platform.

- Tasks:

1. Use a dataset with at least 5 numerical variables.
2. Calculate the correlation matrix using df.corr().
3. Visualization: Use plt.imshow() or sns.heatmap() to visualize the correlations.
4. Requirement: Use a diverging color map (e.g., 'RdBu') to clearly distinguish between positive and negative correlations.

Ans :

Code :

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
import numpy as np

# ------------------------------
# Task 1: Create dataset (5+ numerical columns)
# ------------------------------
np.random.seed(0)

df = pd.DataFrame({
    'A': np.random.rand(100),
    'B': np.random.rand(100),
    'C': np.random.rand(100),
    'D': np.random.rand(100),
    'E': np.random.rand(100)
})

print("Dataset Preview:")
print(df.head())

# ------------------------------
# Task 2: Correlation Matrix
# ------------------------------
corr = df.corr()

print("\nCorrelation Matrix:")
print(corr)

# ------------------------------
# Task 3 & 4: Heatmap Visualization
# ------------------------------
plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap='RdBu')

plt.title("Correlation Heatmap")

# Save output (IMPORTANT for WSL)
plt.savefig("heatmap.png")

print("\nHeatmap saved successfully!")
```

user@DESKTOP-0OTJMH0: ~/

```
  GNU nano 7.2                                  exercise4.py
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

# -------------------------------
# Task 1: Create dataset (5+ numerical columns)
# -------------------------------
np.random.seed(0)

df = pd.DataFrame({
    'A': np.random.rand(100),
    'B': np.random.rand(100),
    'C': np.random.rand(100),
    'D': np.random.rand(100),
    'E': np.random.rand(100)
})

print("Dataset Preview:")
print(df.head())

# -------------------------------
# Task 2: Correlation Matrix
# -------------------------------
corr = df.corr()
```

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corr = df.corr()

print("\nCorrelation Matrix:")
print(corr)

# -------------------------------
# Task 3 & 4: Heatmap Visualization
# -------------------------------
plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap='RdBu')

plt.title("Correlation Heatmap")

# Save output (IMPORTANT for WSL)
plt.savefig("heatmap.png")

print("\nHeatmap saved successfully!")
```

Output :

```
(myenv) user@DESKTOP-0OTJMH0:~/bda_lab/Lab_10$ python3 exercise4.py
Dataset Preview:
          A         B         C         D         E
0  0.548814  0.677817  0.311796  0.906555  0.401260
1  0.715189  0.270008  0.696343  0.774047  0.929291
2  0.602763  0.735194  0.377752  0.333145  0.099615
3  0.544883  0.962189  0.179604  0.081101  0.945302
4  0.423655  0.248753  0.024679  0.407241  0.869489

Correlation Matrix:
          A         B         C         D         E
A  1.000000 -0.066107 -0.036512  0.021895 -0.005389
B -0.066107  1.000000 -0.124047 -0.061987 -0.114401
C -0.036512 -0.124047  1.000000 -0.115117  0.062719
D  0.021895 -0.061987 -0.115117  1.000000  0.033356
E -0.005389 -0.114401  0.062719  0.033356  1.000000

Heatmap saved successfully!
(myenv) user@DESKTOP-0OTJMH0:~/bda_lab/Lab_10$
```